LMS-Based Identification and Compensation of Timing Mismatches in a Two-Channel Time-Interleaved Analog-to-Digital Converter

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***Abstract*— A time-interleaved ADC (TIADC) increases the overall sampling rate by combining multiple slow ADCs. However, the performance of a TIADC suffers from several mismatches such as time, offset, and gain mismatches. This paper deals with the identification and compensation of timing mismatches in a TIADC using the least mean square (LMS) algorithm. The method only requires a bandlimited and oversampled input signal. We present a detailed discussion and demonstrate the effectiveness of the proposed method by numerical simulations.**

I. Introduction

A time-interleaved analog-to-digital converter (TIADC) has M parallel channel ADCs, where each channel ADC has a sampling rate of f_s/M [1]. The overall system is operating with a sampling rate of f_s. In terms of the timing of a TIADC, a sample is taken by another channel ADC at each time step and a digital output is produced. Hence each channel ADC has a sampling period of MT_s and the overall time-interleaved system has a period of T_s. However, the performance of a time-interleaved architecture is limited by the mismatches, such as gain, offset, and timing mismatches. Among these three types of mismatches, timing mismatches are most difficult to identify and correct.

Different algorithms have been developed for timing mismatch correction [2], [3] and other works have proposed adaptive techniques for the identification and correction of timing mismatches [4], [5]-[6]. In [5]-[6] the authors have proposed an adaptive filter structure based on the null-steering filter bank and have used a stochastic gradient descent adaptive algorithm for the timing mismatch estimation.

In this paper we introduce a method for the identification and compensation of timing mismatches (Fig. 1) based on a simple adaptive technique comparable to the one given in [5]-[6]. We use high-pass filters, a differentiator, and an adaptive multiplier in our identification structure for the estimation of timing mismatches. For the derivations in this paper, we assume that offset and gain mismatches have already been corrected.

The outline of this paper is as follows: In Section 2, we present the system model, Section 3 discusses the details of the identification structure, Section 4 explains the compensation of error spectra in the output of a TIADC and Section 5 shows the simulation results.

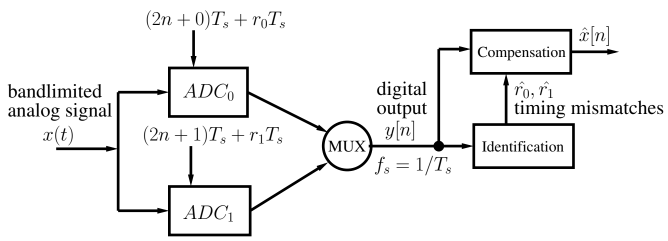

Fig. 1. Identification and compensation of timing mismatches with a bandlimited oversampled input signal for a two-channel TIADC.

II. System Model

The output spectrum of a TIADC is shown in Fig. 2, where the bandlimited input signal $X(j\Omega)$ is oversampled by some oversampling ratio $K = \frac{\Omega_s}{2\Omega_b}$ [7]. To be specific, the input signal is bandlimited, i.e., $X(j\Omega) = 0$ for $|\Omega| > \Omega_b$ and $\Omega_b T_s < \pi$. For bandlimited input signals, the output of a two-channel TIADC with timing mismatches can be represented as [8]

$$Y(e^{j\omega}) = \sum_{k=0}^{1} \alpha_k(e^{j(\omega-k\pi)})X(e^{j(\omega-k\pi)}) \tag{1}$$

where $\omega = \Omega T_s$. The coefficients $\alpha_k(e^{j\omega})$ are given by

$$\alpha_k(e^{j\omega}) = \frac{1}{2}\sum_{m=0}^{1} e^{H_d(e^{j\omega})r_m} e^{-jkm\pi} \tag{2}$$

where r_m are relative time errors that account for the deviations r_mT_s from the ideal sampling instants nT_s leading to the timing mismatches in the TIADC as shown in Fig. 3. Furthermore, we have in (2)

$$H_d(e^{j\omega}) = j\omega, -\pi \leq \omega < \pi \tag{3}$$

which is the frequency response of the ideal discrete-time differentiator [9]. Applying Taylor's series expansion

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to $e^{H_d(e^{j\omega})r_m}$ in (2) and neglecting higher order terms yields [8]

$$e^{H_d(e^{j\omega})r_m} \approx 1 + H_d(e^{j\omega})r_m. \tag{4}$$

Using (4), the coefficients α_k in (2) can be evaluated as

$$\alpha_0(e^{j\omega}) = 1 + H_d(e^{j\omega})\frac{r_0}{2} + H_d(e^{j\omega})\frac{r_1}{2} \tag{5}$$

$$\alpha_1(e^{j\omega}) = H_d(e^{j\omega})\frac{r_0}{2} - H_d(e^{j\omega})\frac{r_1}{2}. \tag{6}$$

Now replacing $\alpha_0(e^{j\omega})$ and $\alpha_1(e^{j\omega})$ in (1) by (5) and (6), respectively, gives [8]

$$Y(e^{j\omega}) \approx X(e^{j\omega}) + E(e^{j\omega}). \tag{7}$$

From (7), it can be seen that the output spectrum of the TIADC is composed of the original signal spectrum $X(e^{j\omega})$ and the error spectrum $E(e^{j\omega})$ that is given by

$$\begin{aligned} E(e^{j\omega}) =& H_d(e^{j\omega})X(e^{j\omega})R_0 \\ &+ H_d(e^{j(\omega-\pi)})X(e^{j(\omega-\pi)})R_1 \end{aligned} \tag{8}$$

where $R_0 = \frac{r_0+r_1}{2}$ is the overall delay of the two-channel TIADC and $R_1 = \frac{r_0-r_1}{2}$ is the weighted difference of the timing errors r_0 and r_1. Since we are only interested in the mismatches, we can without loss of generality assume that $R_0 = 0$. This assumption leads to

$$r_1 = -r_0 \tag{9}$$

and consequently

$$R_1 = \frac{r_0 - (-r_0)}{2} = r_0. \tag{10}$$

Now putting $R_0 = 0$ and the value of R_1 from (10) into (8), the expression for the error spectrum reduces to

$$E(e^{j\omega}) = H_d(e^{j(\omega-\pi)}) \cdot X(e^{j(\omega-\pi)}) \cdot r_0. \tag{11}$$

So the error spectrum depends on the time error r_0 and the differentiated input signal $x[n]$. Substituting $E(e^{j\omega})$ from (11) into (7) and applying the inverse discrete-time Fourier transform (DTFT) gives the output $y[n]$ of the two-channel TIADC

$$y[n] = x[n] + e[n] \tag{12}$$

with

$$e[n] = (-1)^n \cdot (x * h_d)[n] \cdot r_0. \tag{13}$$

III. Identification

Based on the system model from the previous section, we now introduce the identification structure shown in Fig. 4. An oversampling of the input signal leads to a region where no signal power is present [5]-[6], [7]. However, aliased spectral components that are due to timing mismatches appear (c.f. Fig. 2). In order to identify the timing mismatches, the aliased spectral components have to contribute significant error power into the mismatch band. The identification structure needs to correctly estimate the timing error r_0 and based on

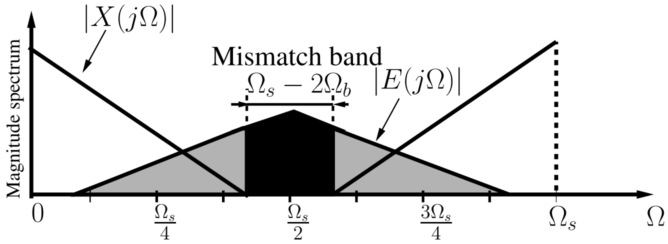

Fig. 2. The output spectrum $|Y(j\Omega)|$ of a two-channel ($M = 2$) TIADC with input spectra $|X(j\Omega)|$, and error spectra $|E(j\Omega)|$. The band where all spectral components except the input spectrum itself contribute is called mismatch band (black filled).

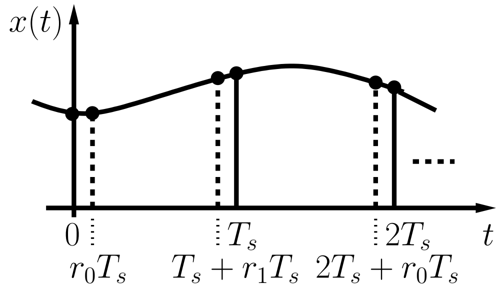

Fig. 3. Time errors (dashed) in a two-channel TIADC are deviations from the ideal sampling instants nT_s (solid).

this estimated value, the error $e[n]$ in the output $y[n]$ of a TIADC can be compensated.

In Fig. 4 the frequency response of the differentiator is approximated by the linear-phase FIR filter $h_d[n]$. There are two high-pass filters having identical impulse responses $f[n]$. They pass the signal only inside the mismatch band. The output $\epsilon[n]$ of the identification structure is given by

$$\epsilon[n] = \hat{y}[n] - \bar{e}[n]. \tag{14}$$

The signal $\hat{y}[n]$ is given by

$$\hat{y}[n] = (y * f)[n]. \tag{15}$$

Substituting (12) into (15) gives

$$\hat{y}[n] = (x * f)[n] + (e * f)[n]. \tag{16}$$

Since the high-pass filter $f[n]$ attenuates the input signal band (c.f Fig. 2), we can write

$$\hat{y}[n] = (e * f)[n]. \tag{17}$$

Substituting $e[n]$ from (13) in (17), $\hat{y}[n]$ becomes

$$\hat{y}[n] = \Big(\Big[(-1)^n \cdot (x * h_d)[n]\Big] * f\Big)[n] \cdot r_0. \tag{18}$$

The estimated error signal $\bar{e}[n]$ in the mismatch band is given by

$$\bar{e}[n] = \bar{y}[n] \cdot \hat{r}_0[n]. \tag{19}$$

The term $\hat{r}_0[n]$ in (19) represents the time-varying coefficient of the adaptive multiplier that corresponds to the estimate of the time error r_0, while $\bar{y}[n]$ is given by

$$\bar{y}[n] = \Big(\Big[(-1)^n \cdot (y * h_d)[n]\Big] * f\Big)[n]. \tag{20}$$

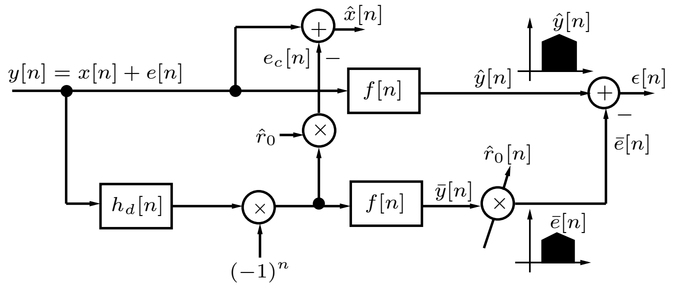

Fig. 4. Identification and compensation structure with a differentiator $h_d[n]$, high-pass filters $f[n]$, and an adaptive multiplier $\hat{r}_0[n]$.

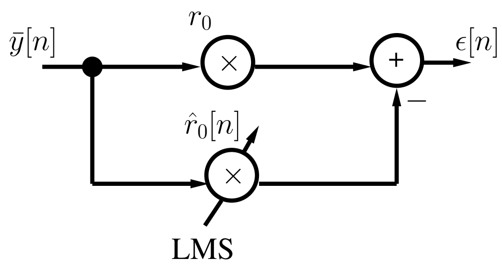

Fig. 5. Identification of the time error r_0. The time-varying coefficient $\hat{r}_0[n]$ is adapted using the LMS algorithm in order to estimate the value of r_0.

As for small timing mismatches, the input signal $x[n]$ has a much larger energy than $e[n]$ [10], we can substitute (12) in (20) and neglect $e[n]$, which gives

$$\bar{y}[n] \approx \Big(\Big[(-1)^n \cdot (x * h_d)[n]\Big] * f\Big)[n]. \tag{21}$$

Now substituting $\hat{y}[n]$ and $\bar{e}[n]$ from (18) and (19) respectively in (14), we can write $\epsilon[n]$ as

$$\epsilon[n] = \Big(\Big[(-1)^n \cdot (x * h_d)[n]\Big] * f\Big)[n] \cdot (r_0 - \hat{r}_0[n]) \tag{22}$$

or by using (21) as

$$\epsilon[n] = \bar{y}[n] \cdot (r_0 - \hat{r}_0[n]). \tag{23}$$

Hence the expression for $\epsilon[n]$ reduces to a simple system identification problem as shown in Fig. 5. To estimate the value of the timing error r_0, the coefficient $\hat{r}_0[n]$ is adapted using an LMS algorithm [11], i.e.,

$$\hat{r}_0[n] = \hat{r}_0[n-1] + \mu \cdot \bar{y}[n] \cdot \epsilon[n]. \tag{24}$$

IV. Compensation

The basic idea of compensation is to cancel the error $e[n]$ in the output $y[n]$ of the TIADC and has already been introduced in [8]. The error $e[n]$ can be compensated by subtracting an error compensation sequence $e_c[n]$ from $y[n]$. This will result in a reconstructed input signal $\hat{x}[n]$ given by

$$\hat{x}[n] = y[n] - e_c[n]. \tag{25}$$

The error compensation sequence $e_c[n]$ as shown in Fig. 4 can be written as

$$e_c[n] = (-1)^n \cdot (y * h_d)[n] \cdot \hat{r}_0. \tag{26}$$

Making the same assumption as in (21), we can rewrite (26) as

$$e_c[n] \approx (-1)^n \cdot (x * h_d)[n] \cdot \hat{r}_0. \tag{27}$$

Substituting (12) and (27) in (25) and rearranging gives

$$\hat{x}[n] = x[n] + \Big((-1)^n \cdot (x * h_d)\Big)[n] \cdot (r_0 - \hat{r}_0). \tag{28}$$

The second term in (28) corresponds to the estimated error spectra present in the output of the TIADC and can be cancelled if $\hat{r}_0$ converges to r_0, leading us to

$$\hat{x}[n] \approx x[n]. \tag{29}$$

V. Simulation Results

This section presents the simulation results that illustrate the performance of the proposed method. For the simulations, we have ignored the quantization effects. The input signal has been white-Gaussian noise (WGN) with variance $\sigma = 1$ and from this input signal we have taken 524288 samples. The value of the step size μ was equal to 10^{-3}. The time error r_0 was -0.02 and the time error r_1 was $+0.02$. The differentiator $h_d[n]$ and the high-pass filters $f[n]$ have been designed using the Matlab function 'firpm'. The corner frequency of $f[n]$ was 0.77π and the number of taps of $h_d[n]$ were 11. The overall compensation performance has been evaluated by computing the signal-to-noise-and-distortion ratio (SINAD) [12] over a different number of samples. The value of the SINAD before identification and compensation is given by

$$\text{SINAD} = 10\log_{10}\left(\frac{\sum_{n=0}^{N-1} |x[n]|^2}{\sum_{n=0}^{N-1} |x[n] - y[n]|^2}\right) \tag{30}$$

and after identification and compensation is given by

$$\text{SINAD} = 10\log_{10}\left(\frac{\sum_{n=0}^{N-1} |x[n]|^2}{\sum_{n=0}^{N-1} |x[n] - \hat{x}[n]|^2}\right). \tag{31}$$

Figure 6 shows the power spectrum of the output signal $y[n]$ for a bandlimited WGN input over the band $[-0.7\pi, 0.7\pi]$ before identification and compensation of timing mismatches. The calculated value of SINAD is approximately 26 dB. The convergence of timing mismatches with an increasing number of samples is shown in Fig. 7. The timing error r_0 has converged to its correct value approximately after 6×10^4 samples. Figure 8 shows the output power spectrum of $\hat{x}[n]$ after timing mismatch identification. As can be seen, the error spectra in the mismatch band have very little power. Now the calculated value of the SINAD is approximately 59 dB leading to an improvement of 33 dB.

VI. Conclusions

In this paper we have proposed a method for the identification and compensation of timing mismatches in a two-channel time-interleaved ADC using an adaptive approach based on an LMS algorithm. We have assumed a bandlimited and oversampled input signal that leads to the creation of a mismatch band containing only aliased spectral components.

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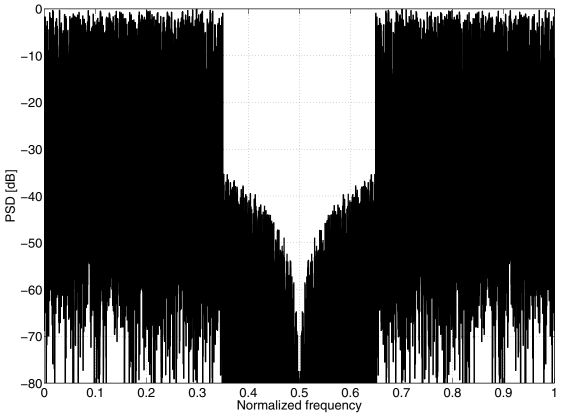

Fig. 6. The power spectrum of the TIADC output $y[n]$. The calculated value of the SINAD is approximately 26 dB.

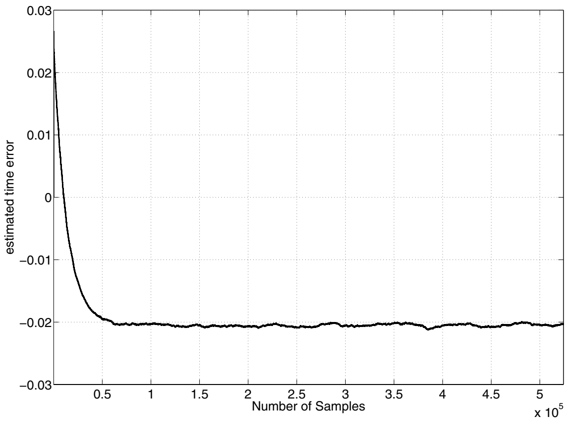

Fig. 7. Convergence of the estimated timing error $\hat{r}_0[n]$ for a TIADC with $r_0 = -0.02$ and $r_1 = +0.02$, a bandlimited WGN input for 524288 samples, and $\mu = 10^{-3}$.

The timing mismatches have been identified by minimizing the energy in this mismatch band. Based on the identified value of timing mismatches, the error in the output of a TIADC has been compensated. Hence by using simple high-pass filters, a differentiator, and an adaptive multiplier, we can identify and compensate the timing mismatches in a TIADC.

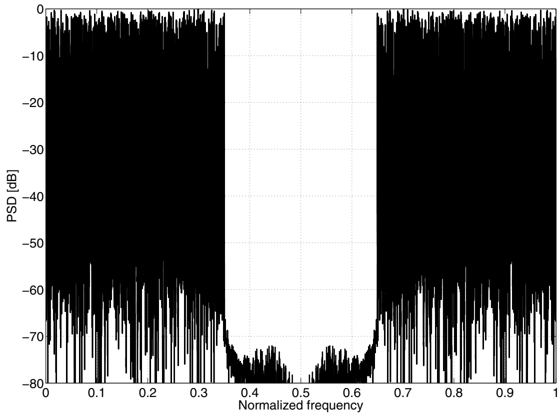

Fig. 8. The power spectrum of the compensated output $\hat{x}[n]$ after the identification process has been converged. The calculated value of the SINAD is approximately 59 dB.

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